

■ ■ Multi-ESP32 MQTT Drone Control System

Communicating with a Drone using Python and MQTT

■ Project Information

Project Title: Multi-ESP32 MQTT Drone Control System

Developed For: Meritus.AI

Developed By: Apurva Donde

Role: Drone Systems Developer | Tools & Automation | STEM Education

■ Overview

This project establishes a multi-layer MQTT communication system between a Python script, two ESP32 boards, and a Pluto Drone. It demonstrates how commands can be transmitted over Wi-Fi using MQTT topics to control drone operations such as arm, takeoff, land, and disarm.

■ Hardware Requirements

Component	Quantity	Description
ESP32	2	Wi-Fi enabled microcontrollers
Pluto Drone	1	Target drone to receive and execute commands

■ System Topology

Python Script → ESP32 (Bridge / Hub) → ESP32 (Listener) → Pluto Drone
(MQTT) (Wi-Fi) (Serial / Command Execution)

■ ■ Components Description

1. ESP32 – MQTT Bridge (Wi-Fi Hub)

- Acts as a Wi-Fi Access Point (AP).
- Subscribes to topic: pluto/cmd/#
- Forwards messages to: bridge/esp32/cmd
- Displays logs via Serial Monitor.

2. ESP32 – MQTT Listener

- Connects to the ESP32 Hub Wi-Fi network.
- Subscribes to topic: bridge/esp32/cmd
- Executes commands: arm, disarm, takeoff, land.

3. Python – Command Publisher

- Script: send_mqtt_cmd.py
- Publishes messages to: pluto/cmd/arm, pluto/cmd/disarm, pluto/cmd/takeoff, pluto/cmd/land.
- Run command: python3 send_mqtt_cmd.py arm

■ Command Execution via Terminal

```
mosquitto_pub -h 192.168.4.2 -t 'pluto/cmd/arm' -m 'arm'
mosquitto_pub -h 192.168.4.2 -t 'pluto/cmd/takeoff' -m 'takeoff'
mosquitto_pub -h 192.168.4.2 -t 'pluto/cmd/land' -m 'land'
```

■ Troubleshooting & Network Utilities

Check Local IP Address:

```
ipconfig getifaddr en0
```

Issue	Possible Cause	Solution
MQTT messages not received	Incorrect topic or broker IP	Verify MQTT topics and broker IP
ESP32 not connecting	Wrong SSID/password	Double-check Wi-Fi credentials
No serial output	Baud rate mismatch	Ensure Serial Monitor is set to correct baud rate

■ **Future Improvements**

- Add telemetry feedback from the drone to the Python dashboard.
- Implement command acknowledgment for improved reliability.

■■■ **Author**

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